

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
2nd Semester of B.Pharm. Examination
University Theory Examination May 2017
MA141 Advanced Mathematics

Date: 17.05.2017, Wednesday Time: 10:00 a.m. to 01:00 p.m. Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.
4. Use of scientific calculator is allowed.

SECTION- I**Q 1** Attempt any TWO of the following

- A** (I) Verify Lagrange's mean value theorem for $f(x) = x^3 + 5$ where $x \in [-1, 3]$. **05**
- (II) Show that $2xydx + (x^2 - y^2)dy = 0$ is an exact differential equation and obtain general solution of it. **05**
- B** (I) Verify Cauchy's mean value theorem for $f(x) = x^3$ and $g(x) = 2 - x$ where $x \in [0, 2]$. **05**
- (II) Find general solution of $y'' + y' + 2y = 0$. **05**
- C** (I) Identify the type of an ordinary differential equation $\frac{dy}{dx} + y = e^{-x}$ and find particular solution with help of initial condition $y(0) = 1$. **05**
- (II) Solve differential equation $\frac{dy}{dx} = \frac{x^3 + 2xy^2}{x^2y}$. **05**

Q 2 Attempt any TWO of the following

- A** (I) Find Maclaurin series of e^x and $\cos x$. **05**
- (II) Identify the type of an ordinary differential equation $\frac{dy}{dx} + 4\frac{y}{x} = x^3$ and find particular solution with help of initial condition $y(0) = 0$. **05**
- B** (I) Find Taylor series of function $f(x) = x^2 + 2x + 1$ around $x = 3$. Also show that Maclaurin series is a particular case of Taylor series. **05**
- (II) Find general solution of $y'' + 2y' + y = 0$. **05**
- C** (I) Solve $y'' - 5y' + 6y = e^{3x}$. **05**
- (II) Obtain solution of $x dx + y^2 dy = 0$ and $y'' - y = 0$. **05**

SECTION- II

Q 3 Attempt any **TWO** of the following

- A (I) The population of a community is known to increase at a rate proportional to the number of people present at a time t . If the population has doubled in 6 years, how long it will take to triple? 05
- (II) Write statement of first shifting theorem and using it find Laplace transform of $t^2 e^{3t}$ and $e^t \cos 5t$. 05
- B (I) Find n^{th} derivative of $y = \log\left(\frac{1+x}{1-x}\right) + 2^x + 8$. 05
- (II) Evaluate $L(\sin 3t - t^3 + e^{-3t})$ and $L\left[(2t+5)^2 + 2e^{-t} + 5\right]$. 05
- C (I) Write statement of Leibnitz's theorem for n^{th} derivative of product of two functions and using it find n^{th} derivative of $y = x^2 3^{5x}$. 05
- (II) Differential equation $\frac{ds}{dt} + 4s = e^t$ represents a relation between sugar(s) level in blood and time(t) of a person. Solve the differential equation for sugar(s) level in term of t . 05

Q 4 Attempt any **TWO** of the following

- A (I) Find 30^{th} derivative of $y = \sin(x-1) + 2\cos(2x+1) + 3e^x$. 05
- (II) If the rate of change of glucose is propotinal to time and at 1 hour rate is given by 300 mlgram per hour, then find glucoze at $t = 5$ hour where at intial stage glucose in blood is zero. 05
- B (I) Solve the differential equation $y'' + y = 0$ with intial condition $y(0) = 1$ and $y'(0) = 0$ using Laplace transform. 05
- (II) If rate of change of population is propotinal to time and at 1 hour rate is given by 100 person per hour, then find number of person at $t = 3$ hour where at intial stage number of person is zero. 05
- C (I) Find Laplace transform of e^{5t} , $\sin 6t$ and $\cos t$. Find formula for n^{th} derivative of $y = 3^x$. 05
- (II) Bacteria multiply at the rate proportional to the number present. If the original number N doubles in three hours, then find required time so that the number of bacteria will be $4N$. 05